Search Interest Predicts Future Duolingo Growth

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Outline:

- Motivation & Hypothesis
- Research Question, Modeling Approach, and Goal
- Data Explanation and Acquisition
- Analysis Plan and Justification
- Tricky Analysis Decision
- Bias and Uncertainty Validation
- Results and Conclusions
- Next Steps
- References and Resources

Motivation:

- Understand trends and relationships in Duolingo's growth using time-series data
- Companies want to predict growth before it happens
- Search data could predict performance

Applications:

- Provide insights on factors related to user growth and subscribers
- Inform strategies to reduce churn and improve monetization

Hypothesis: Increases in Google search interest for Duolingo are positively and significantly associated with future increases in daily active users, paid subscribers, and revenue, indicating that search interest functions as a leading indicator of company growth.

Research question:
To what extent does Google search interest in Duolingo predict changes in daily active users, paid subscribers, and revenue over subsequent quarters, and does a lagged relationship outperform a contemporaneous one?

Modeling Approach:

- Perform preliminary trend analysis
- Compute Pearson and Spearman correlations
- Analyze lagged relationships (T → T+1, T+2)
- Visualize relationships using scatterplots + trendlines

Goal: Identify the optimal lead-time relationship between Google search interest and Duolingo's performance metrics, specifically determining if a 1-quarter or 2-quarter lag provides the most statistically significant predictive power for DAU, subscribers, and revenue.

Data Acquisition and Explanation of Google Trend Data

Column	Description
Period	Year the data is from 2022-2025 with Quarter
duolingo	% change of google queries about 'Duolingo'
duolingo spanish	% change of google queries about 'duolingo spanish'
what is duolingo	% change of google queries about 'what is duolingo'
duolingo english	% change of google queries about 'duolingo english'
language learning app	% change of google queries about 'language learning app'

Summary of Dataset:

- 6 columns including:
  - Time
  - 5 different Duolingo related search terms
- 17 rows of data
- Origin: Google Trend website

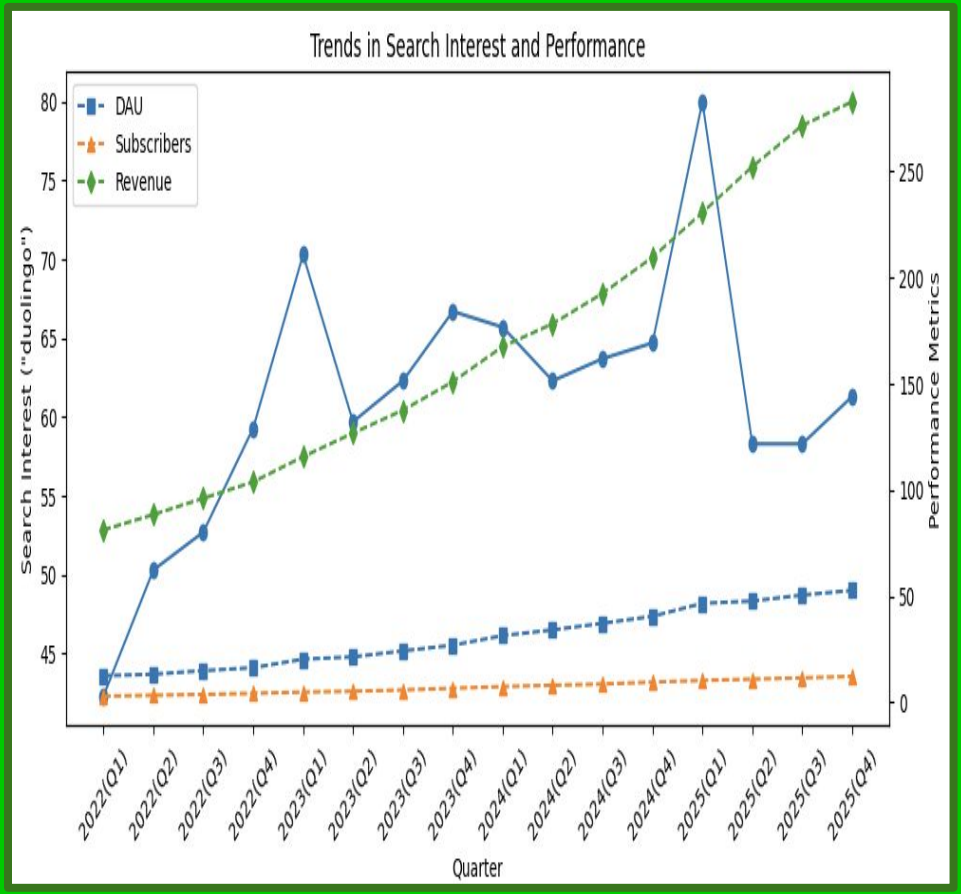
Data Acquisition and Explanation of Duolingo Data

Column	Description
Period	Year the data is from 2022-2025 with Quarter
DAU	Daily Active Users per quarter (in millions)
Paid_Subscribers	Paid subscribers at the end of each quarter (in millions)
Revenue	Recognized revenue by quarter (in millions)

Summary of Dataset:

- 5 columns including:
  - Period
  - DAU
  - Paid Subscribers
  - Revenue
- 17 rows of data
- Origin: Company Data

Analysis Plan



Steps:

1) Merge datasets by time period
2) Compute same-quarter correlations
3) Compute lagged correlations (1–2 quarters ahead)
4) Evaluate statistical significance ($p < 0.05$)
5) Visualize relationships with regression lines

Challenges Faced:

Differences in time frequency and scaling between datasets

Tricky Decision Analysis

Decision: Choosing the Optimal Lag Length (1-Quarter vs. 2-Quarter)

- We had to decide whether to treat search interest as a 1-quarter or 2-quarter leading indicator of performance

Why it was Tricky

- A 1-quarter lag is more intuitive, but a 2-quarter lag could reflect slower monetization pipelines
- Using too short a lag risks capturing coincidental co-movement; too long a lag weakens the causal story

Impact of the decision

- The 2-quarter lag produced meaningfully stronger correlations across all three metrics, supporting it as the better choice

Bias and Uncertainty Validation

Biases Considered

- **Search ambiguity:** Google Trends reports relative interest, not absolute query volume
- **Small sample size:** only 17 observations, making results sensitive to outliers
- **Omitted variables:** marketing campaigns likely influence both search and growth but aren't captured

Uncertainty in Results

- Lag 0 correlations did not reach statistical significance ($p > 0.05$) — cannot be distinguished from chance
- Only the 2-quarter lag results are statistically significant ($p < 0.05$)
- Small sample size means correlation estimates carry wide confidence intervals

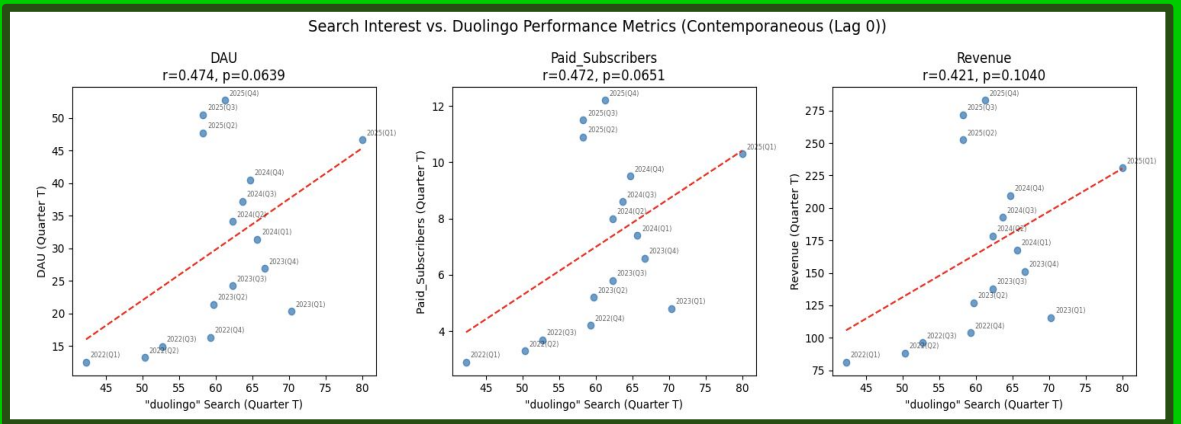
How Bias Was Addressed

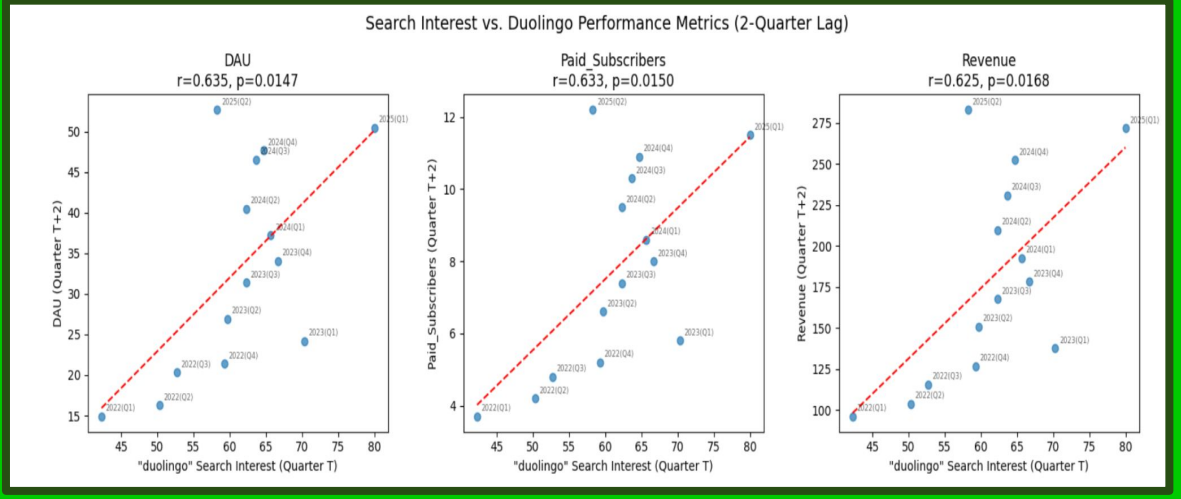
- Directional trends and relative changes rather than point estimates
- Aligned data sets to quarterly time intervals
- Relationships as correlational, not causal

How Uncertainty Was Evaluated

- Compare correlations at Lag 0, Lag 1, and Lag 2
- Reported both Pearson and Spearman correlations to account for potential non-linearity

Results and conclusions





Conclusion:

- Hypothesis is **partially supported**: search interest is positively associated with future Duolingo growth, but the relationship is stronger and only statistically significant when a 2-quarter lag is applied
- These results suggest Google Trends data could serve as a real-time signal for anticipating Duolingo's performance roughly 6 months ahead
- Small sample size warrants caution in generalizing

Next Steps

New Lines of Exploration:

- Use higher-frequency data (monthly instead of quarterly)
- Determine if certain times of year drive stronger user engagement

Improvements:

- Use a longer period of time-series data
- Include more variables such as marketing activity

New Questions:

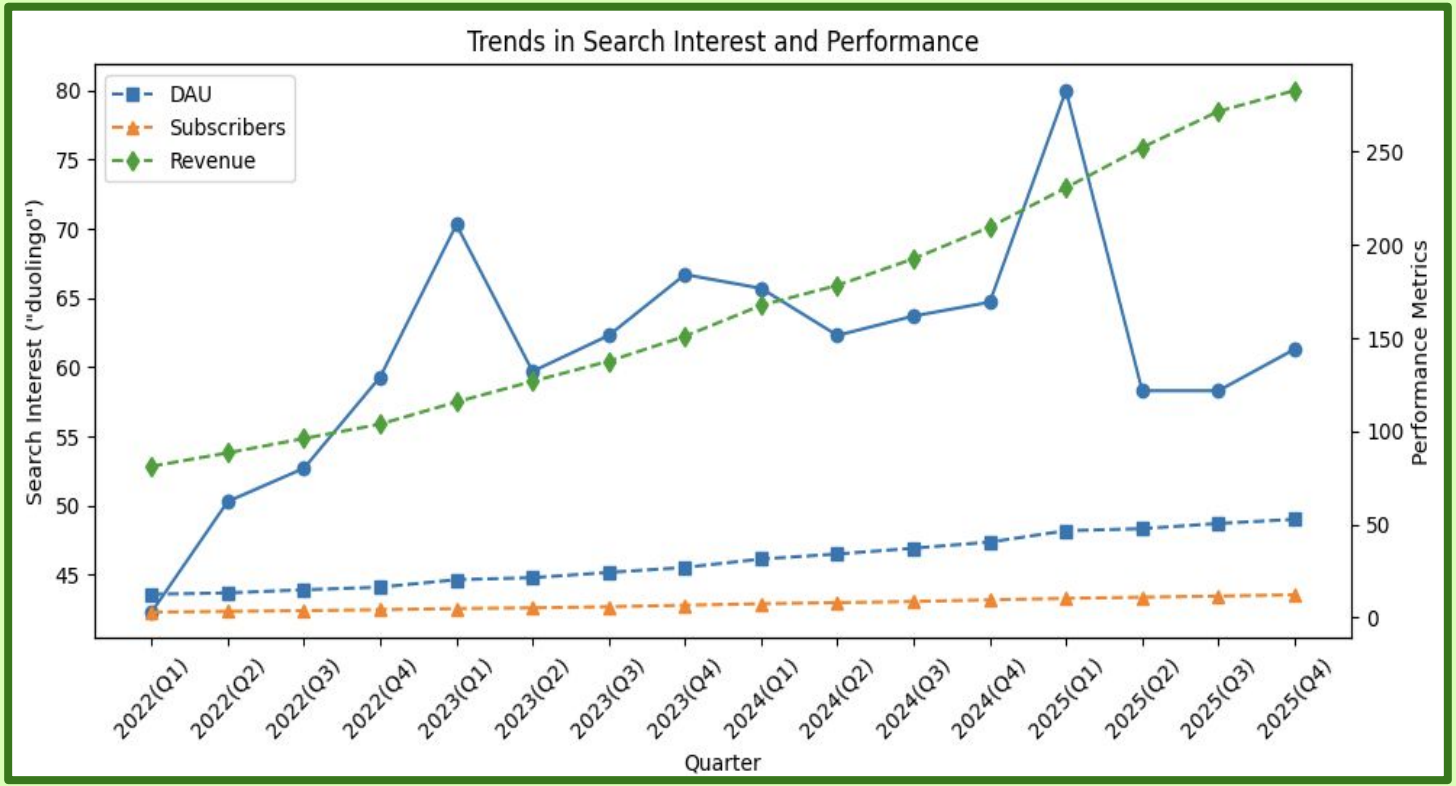
- How often do active users turn into paid subscribers over time?
- How are seasonal changes affected by education schedules, holidays, or marketing campaigns?

References

[1] GeeksforGeeks, “What is a Trend in Time Series?,” *GeeksforGeeks*. [Online]. Available: https://www.geeksforgeeks.org/python/what-is-a-trend-in-time-series/
[2] GeeksforGeeks, “Seasonality Detection in Time Series Data,” *GeeksforGeeks*. [Online]. Available: https://www.geeksforgeeks.org/machine-learning/seasonality-detection-in-time-series-data/
[3] GeeksforGeeks, “How to Calculate Rolling Correlation in Python,” *GeeksforGeeks*. [Online]. Available: https://www.geeksforgeeks.org/python/how-to-calculate-rolling-correlation-in-python/
[4] Investopedia, “Churn Rate: Definition, Formula, and How to Calculate It,” Investopedia. [Online]. Available: https://www.investopedia.com/terms/c/churnrate.asp

Github: https://github.com/dsizzle67/Project2-DS4002

Questions?


Trends in Search Interest and Performance
DAU
Subscribers
Revenue
Search Interest ("duolingo")
Performance Metrics
Quarter
80
75
70
65
60
55
50
45
250
200
150
100
50
0
2022(Q1)
2022(Q2)
2022(Q3)
2022(Q4)
2023(Q1)
2023(Q2)
2023(Q3)
2023(Q4)
2024(Q1)
2024(Q2)
2024(Q3)
2024(Q4)
2025(Q1)
2025(Q2)
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